

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS

1. Smart Campus – IPv4 Subnet Allocation

Given:

IP Address Block = **192.168.100.0/24**

Department	Subnet Mask
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School of Computing	/25
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School of Electronics	/26
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School of Mechanical Engineering	/27
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The subnets are allocated sequentially from **192.168.100.0/24**.

1. Network Address, Broadcast Address, Host Range and Usable Hosts

Department	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
School of Computing (/25)	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics (/26)	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering (/27)	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

2. Calculations

School of Computing – /25

Number of host bits:

$$32 - 25 = 7$$

Total addresses:

$$2^7 = 128$$

Usable host addresses:

$$128 - 2 = 126$$

- Network Address: **192.168.100.0**
- First Host: **192.168.100.1**
- Last Host: **192.168.100.126**
- Broadcast: **192.168.100.127**

School of Electronics – /26

Number of host bits:

$$32 - 26 = 6$$

Total addresses:

$$2^6 = 64$$

Usable host addresses:

$$64 - 2 = 62$$

- Network Address: **192.168.100.128**
- First Host: **192.168.100.129**
- Last Host: **192.168.100.190**
- Broadcast: **192.168.100.191**

School of Mechanical Engineering – /27

Number of host bits:

$$32 - 27 = 5$$

Total addresses:

$$2^5 = 32$$

Usable host addresses:

$$32 - 2 = 30$$

- Network Address: **192.168.100.192**
- First Host: **192.168.100.193**
- Last Host: **192.168.100.222**
- Broadcast: **192.168.100.223**

3. Remaining Unused IP Address Range

The last allocated subnet ends at:

192.168.100.223

Therefore, the remaining unused range is:

192.168.100.224 – 192.168.100.255

This remaining block contains **32 IP addresses**.

Final Answer – Question 1

- **School of Computing:** 192.168.100.0/25 → 126 usable hosts
- **School of Electronics:** 192.168.100.128/26 → 62 usable hosts
- **School of Mechanical Engineering:** 192.168.100.192/27 → 30 usable hosts
- **Unused Range:** 192.168.100.224 – 192.168.100.255

2. Multinational Software Company – IPv4 Subnet Allocation

Given:

IP Address Block = **10.10.0.0/16**

Department	Subnet Mask
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Software Development	/24
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Quality Assurance	/25
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Human Resources	/26
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Executive Management	/27
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The subnets are allocated sequentially.

1. Network Address, Broadcast Address, Host Range and Usable Hosts

Department	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
Software Development (/24)	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance (/25)	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources (/26)	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management (/27)	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

2. Calculations

Software Development – /24

Host bits:

$$32 - 24 = 8$$

Total addresses:

$$2^8 = 256$$

Usable hosts:

$$256 - 2 = 254$$

- Network: **10.10.0.0**
- Host Range: **10.10.0.1 – 10.10.0.254**
- Broadcast: **10.10.0.255**

Quality Assurance – /25

Host bits:

$$32 - 25 = 7$$

Total addresses:

$$2^7 = 128$$

Usable hosts:

$$128 - 2 = 126$$

- Network: **10.10.1.0**
- Host Range: **10.10.1.1 – 10.10.1.126**
- Broadcast: **10.10.1.127**

Human Resources – /26

Host bits:

$$32 - 26 = 6$$

Total addresses:

$$2^6 = 64$$

Usable hosts:

$$64 - 2 = 62$$

- Network: **10.10.1.128**
- Host Range: **10.10.1.129 – 10.10.1.190**
- Broadcast: **10.10.1.191**

Executive Management – /27

Host bits:

$$32 - 27 = 5$$

Total addresses:

$$2^5 = 32$$

Usable hosts:

$$32 - 2 = 30$$

- Network: **10.10.1.192**
- Host Range: **10.10.1.193 – 10.10.1.222**
- Broadcast: **10.10.1.223**

3. Unused IP Address Range

The last allocated subnet ends at:

10.10.1.223

Therefore, the remaining unused IP address range is:

10.10.1.224 – 10.10.255.255

Final Answer – Question 2

- **Software Development:** 10.10.0.0/24 → 254 usable hosts
- **Quality Assurance:** 10.10.1.0/25 → 126 usable hosts
- **Human Resources:** 10.10.1.128/26 → 62 usable hosts
- **Executive Management:** 10.10.1.192/27 → 30 usable hosts
- **Unused Range:** 10.10.1.224 – 10.10.255.255

3. Smart Hospital – IPv6 Addressing

Given IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Notation

The consecutive groups of zeros can be compressed using ::.

Therefore:

Compressed IPv6 Address = 2001:db8:abcd::/64

2. Expanded Full Hexadecimal Representation

The full hexadecimal representation is:

2001:0db8:abcd:0000:0000:0000:0000/64

3. Network Prefix

The prefix length is /64.

Therefore, the first 64 bits represent the network.

Network Prefix:

2001:0db8:abcd:0000::/64

The first four hextets are:

2001 : 0db8 : abcd : 0000

These represent the **64-bit network prefix**.

4. Number of Host Bits

IPv6 addresses contain **128 bits**.

Given prefix length:

64 bits

Therefore:

Host bits = 128 – 64 = 64 bits

5. Total Possible Host Addresses

Number of possible addresses:

$$2^{64}$$

Therefore:

$$2^{64} = 18,446,744,073,709,551,616$$

Final Answer – Question 3

- **Compressed Address:** 2001:db8:abcd::/64
- **Full Address:** 2001:0db8:abcd:0000:0000:0000:0000/64
- **Network Prefix:** 2001:0db8:abcd:0000::/64
- **Host Bits:** 64 bits
- **Total Possible Addresses:** 18,446,744,073,709,551,616

Note: In IPv6, the traditional IPv4 concept of subtracting one network address and one broadcast address is not applied. Therefore, the /64 subnet has **2⁶⁴ possible addresses**.

4. ISP Core Router – Routing Table and Packet Forwarding

Given Routing Table:

Routing Table Entry

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP Address:

192.168.2.45

1. Destination Subnet

The destination IP is:

192.168.2.45

The routing table contains:

192.168.2.0/24

A /24 subnet covers:

192.168.2.0 – 192.168.2.255

Since:

192.168.2.45

falls within this range, the destination subnet is:

192.168.2.0/24

2. Matching Routing Table Entry

The matching routing table entry is:

192.168.2.0/24

3. Next-Hop Network/Interface

The router forwards the packet through the **router interface connected to the 192.168.2.0/24 network**.

Therefore:

Next-hop network/interface → Interface connected to 192.168.2.0/24

The exact interface name or next-hop IP address is **not given in the question**, so it cannot be determined from the provided information.

4. Default Route

If the destination does not match any specific routing table entry, the router uses the default route:

0.0.0.0/0

This represents all IPv4 networks not otherwise specified in the routing table.

Final Answer – Question 4

1. **Destination Subnet:** 192.168.2.0/24
2. **Matching Routing Table Entry:** 192.168.2.0/24
3. **Next-Hop Network/Interface:** Interface connected to 192.168.2.0/24
4. **Default Route:** 0.0.0.0/0

5. Company Office Departments – Layer-3 Router

Given:

Administration LAN

192.168.10.0/26

Gateway:

192.168.10.1

Finance LAN

192.168.10.64/26

Gateway:

192.168.10.65

The subnet mask for /26 is:

255.255.255.192

1. Administration LAN

Network Address

192.168.10.0

Broadcast Address

A /26 subnet contains 64 addresses.

The range is:

192.168.10.0 – 192.168.10.63

Therefore:

- Network Address = **192.168.10.0**
- Broadcast Address = **192.168.10.63**

Valid Host IP Range

192.168.10.1 – 192.168.10.62

Default Gateway

192.168.10.1

Three Suitable Computer IP Addresses

Device	IP Address	Default Gateway
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Administration PC1	192.168.10.2	192.168.10.1
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Administration PC2	192.168.10.3	192.168.10.1
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Administration PC3	192.168.10.4	192.168.10.1
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2. Finance LAN

Network Address

192.168.10.64

Broadcast Address

A /26 subnet contains 64 addresses.

The range is:

192.168.10.64 – 192.168.10.127

Therefore:

- Network Address = **192.168.10.64**
- Broadcast Address = **192.168.10.127**

Valid Host IP Range

192.168.10.65 – 192.168.10.126

Default Gateway

192.168.10.65

Three Suitable Computer IP Addresses

Device	IP Address	Default Gateway
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Finance PC1	192.168.10.66	192.168.10.65
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Finance PC2	192.168.10.67	192.168.10.65
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Finance PC3	192.168.10.68	192.168.10.65
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Question 5

LAN	Network Address	Valid Host Range	Broadcast Address	Default Gateway
Administration	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63	192.168.10.1
Finance	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127	192.168.10.65

Suitable Computer IPs

Administration:

- PC1 → 192.168.10.2
- PC2 → 192.168.10.3
- PC3 → 192.168.10.4
- Gateway → 192.168.10.1

Finance:

- PC1 → 192.168.10.66
- PC2 → 192.168.10.67
- PC3 → 192.168.10.68
- Gateway → 192.168.10.65

